C.R.O.I Technologies

Pulsometer - Final Project Report

Conor Dunne & James Hadley, Jr.
CBE 3300B - May 7th

Who Are We?

Cardiovascular
Rhythmic
Optical
Innovation

Conor Dunne

James Hadley, Jr.

Introduction & Objective

- **Commercial pulsometer**: measure patient's pulse and peripheral blood oxygen saturation
    - Used in medical and at-home care for patients with lung or heart conditions
    - Everyday workout or sleep monitoring

- **Our goal**: create a commercially viable, comfortable, wearable pulsometer to give a real-time display of users' pulse and heart rate in bpm.

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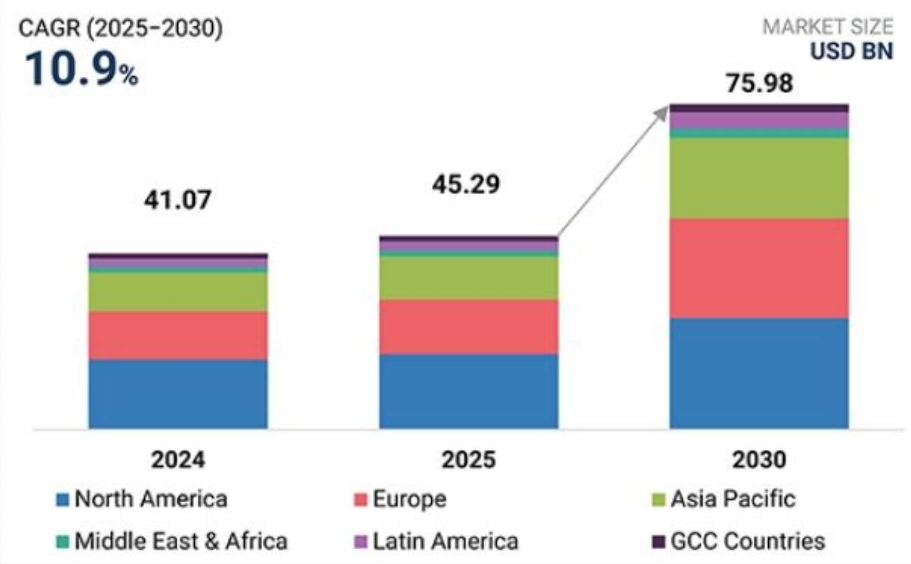
Market & Economic analysis



MarketsandMarkets (2025)

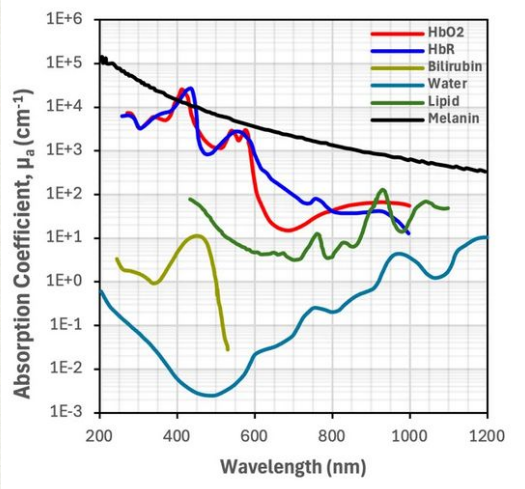
- Wearable Healthcare Devices' Market
    - Trackers, Smartwatches, Patches, & Smart Clothing
    - $45.29 B in 2025
    - $75.98 B in 2030 (67.76% Increase)
    - CAGR of 10.9%
- General Market Analysis
    - Technological advances (AI)
    - Increasing health awareness
- Focus on Pulsometers
    - Potential demand

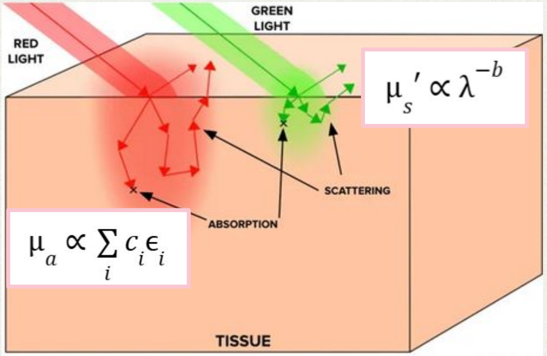
Theory - Photoplethysmography

- **Photoplethysmography (PPG)**: optical technique to measure bioactivity based on the Beer-Lambert law

- Light scattering and absorption dependant on:
    - Wavelength of light
    - Composition / condition of tissue
    - Blood flow

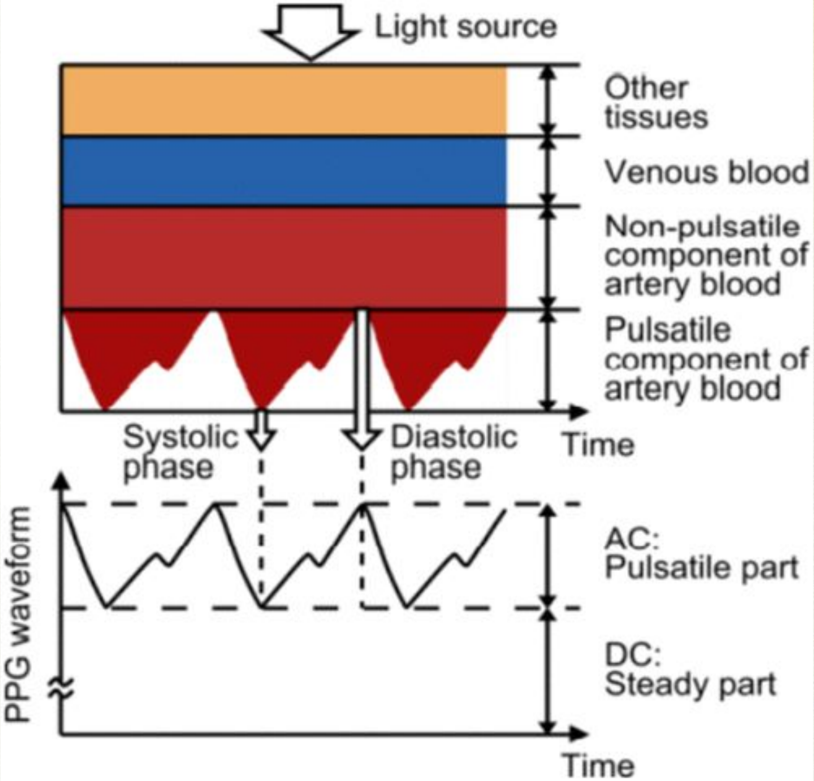
- Measure changes in absorbance from baseline that correspond to light absorbance from flow of oxygenated blood





Engineering Design Constraints

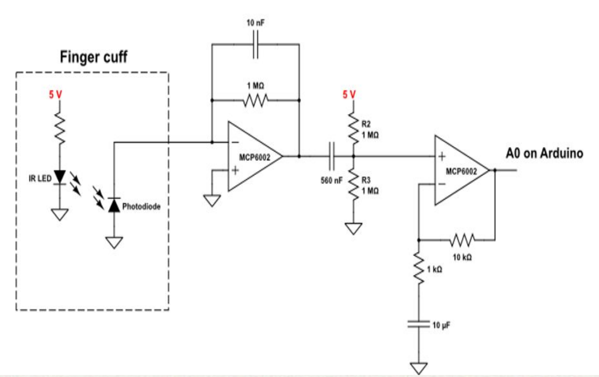
- Signal / noise ratio
  - Signal incredibly small compared to background noise
  - filters and signal processing must be fine tuned
- Environmental / user interactions
  - Ambient room lighting
  - User movement



Methodology

Circuitry

Build the circuit



Code

Code the device to calculate and read BPM

```
void detectPeaks() {
  peakCount = 0;
  acceptedPromCount = 0;
  maxRejectedProm = 0.0;
  int lastPeak = -MIN_DISTANCE;

  for (int i = 1; i < WINDOW - 1; i++) {
    float prev = samples[(sampleIndex + i - 1) % WINDOW];
    float curr = samples[(sampleIndex + i) % WINDOW];
    float next = samples[(sampleIndex + i + 1) % WINDOW];

    if (curr <= prev || curr <= next) continue;
    if (i - lastPeak < MIN_DISTANCE) continue;

    float leftMin = curr;
    for (int j = max(0, i - PROM_LOOKBACK); j < i; j++)
      leftMin = min(leftMin, samples[(sampleIndex + j) % WINDOW]);

    float rightMin = curr;
    for (int j = i + 1; j <= min(WINDOW - 1, i + PROM_LOOKBACK); j++)
      rightMin = min(rightMin, samples[(sampleIndex + j) % WINDOW]);

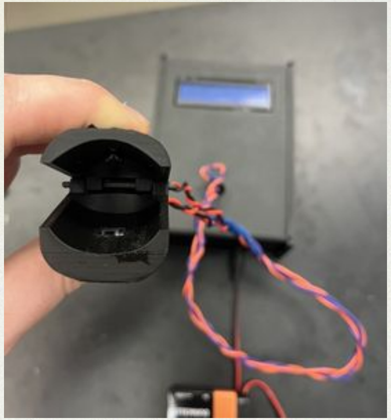
    float prom = curr - max(leftMin, rightMin);

    if (prom < PROMINENCE) {
      if (prom > maxRejectedProm) maxRejectedProm = prom;
      continue;
    }

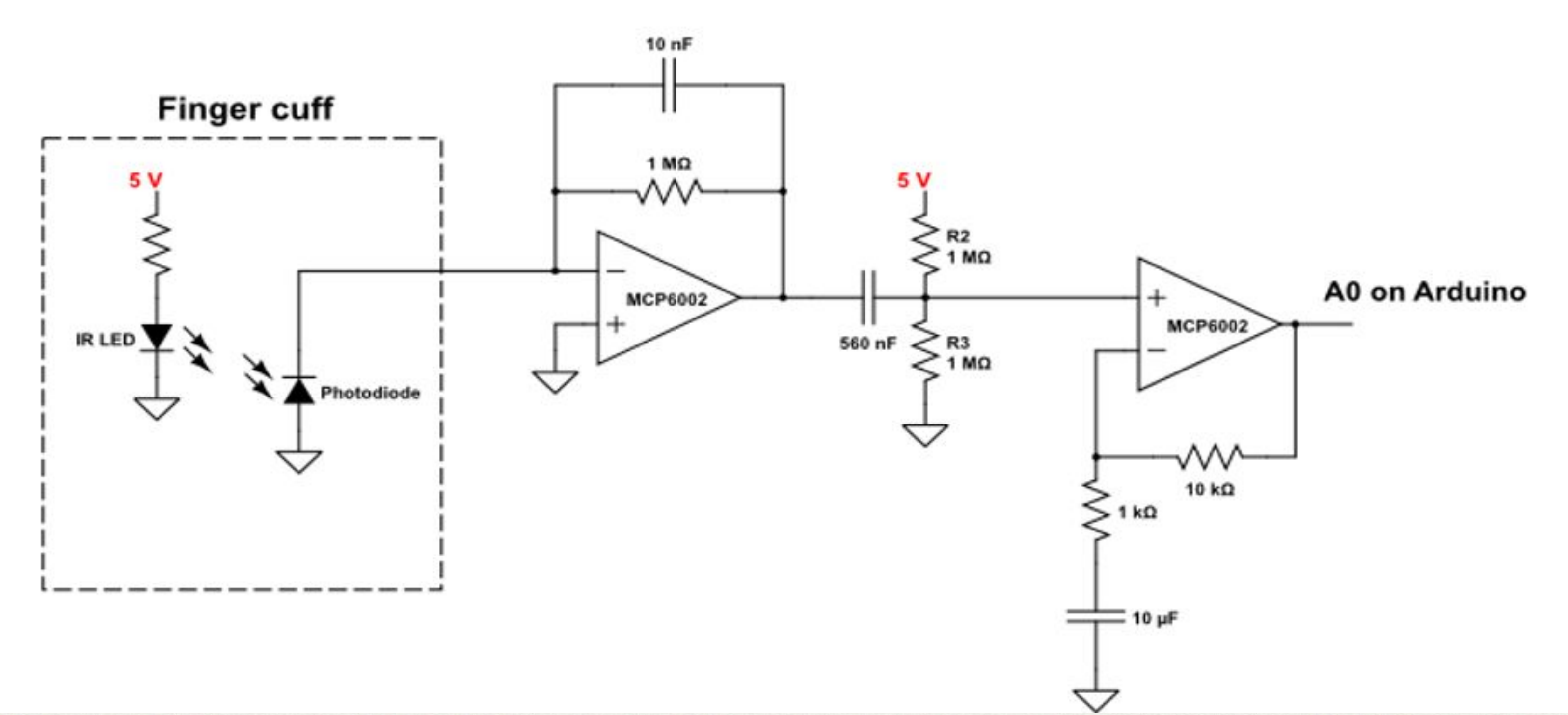
    if (acceptedPromCount < 20) {
      acceptedProms[acceptedPromCount++] = prom;
    }
    peakIndices[peakCount++] = i;
    lastPeak = i;
    if (peakCount >= 20) break;
```

Design

Have the device house the circuits and ergonomic



Electrical Design



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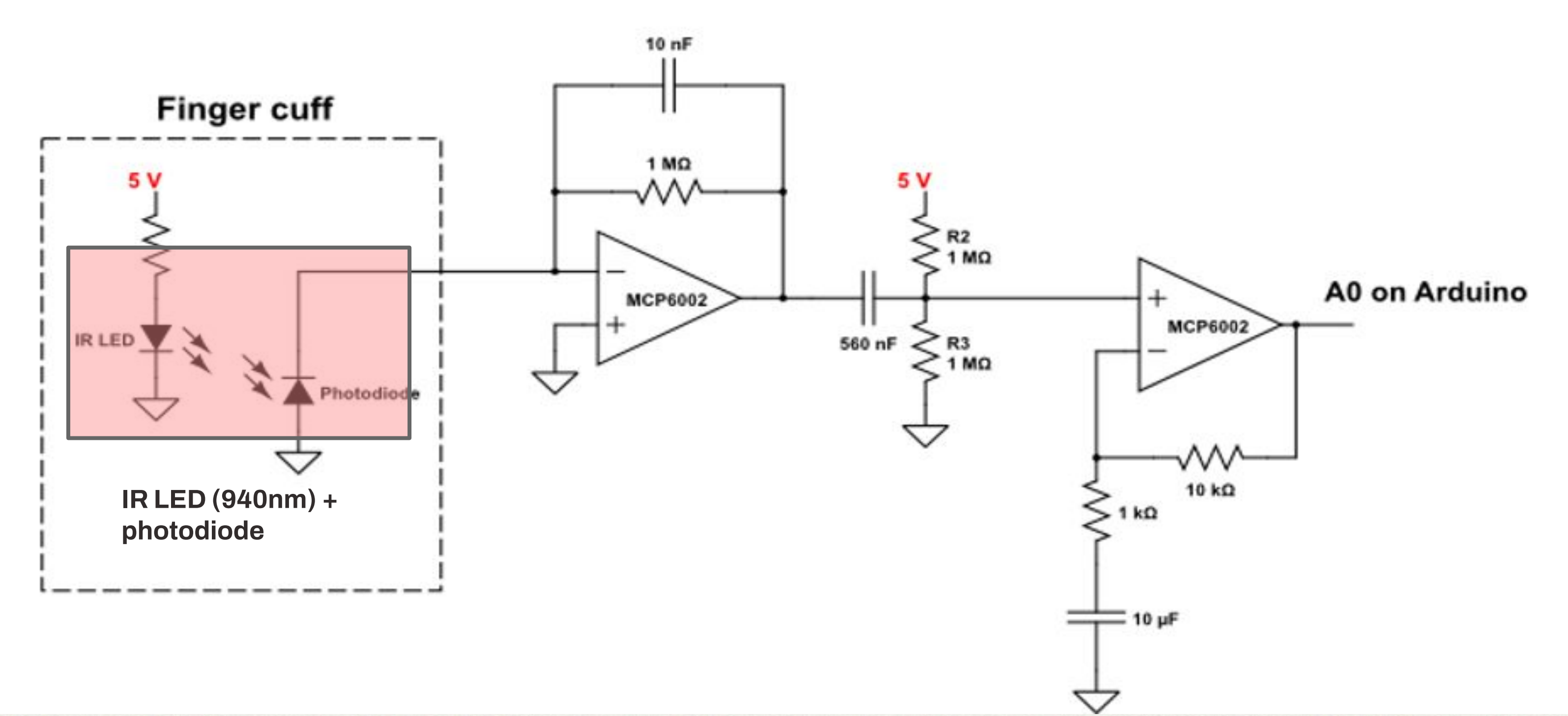

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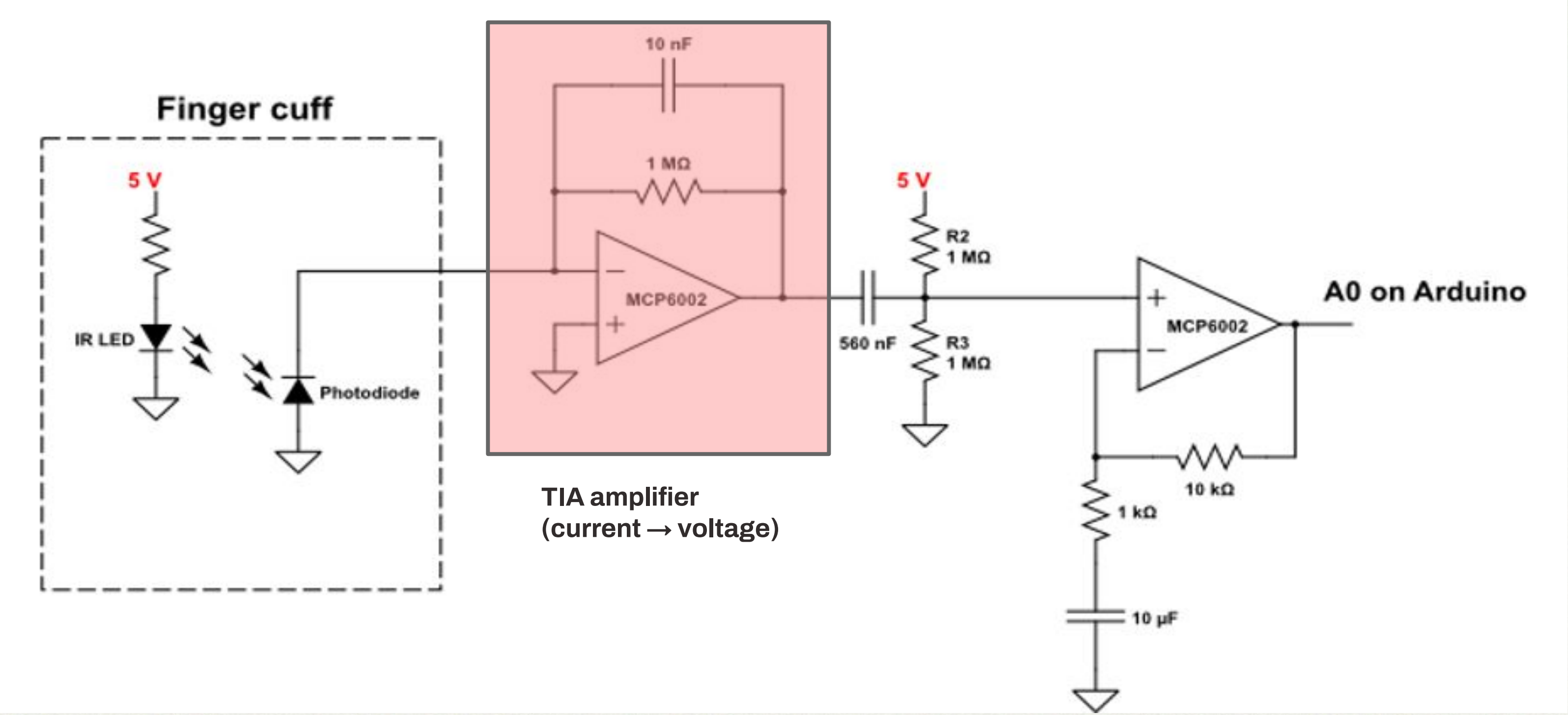

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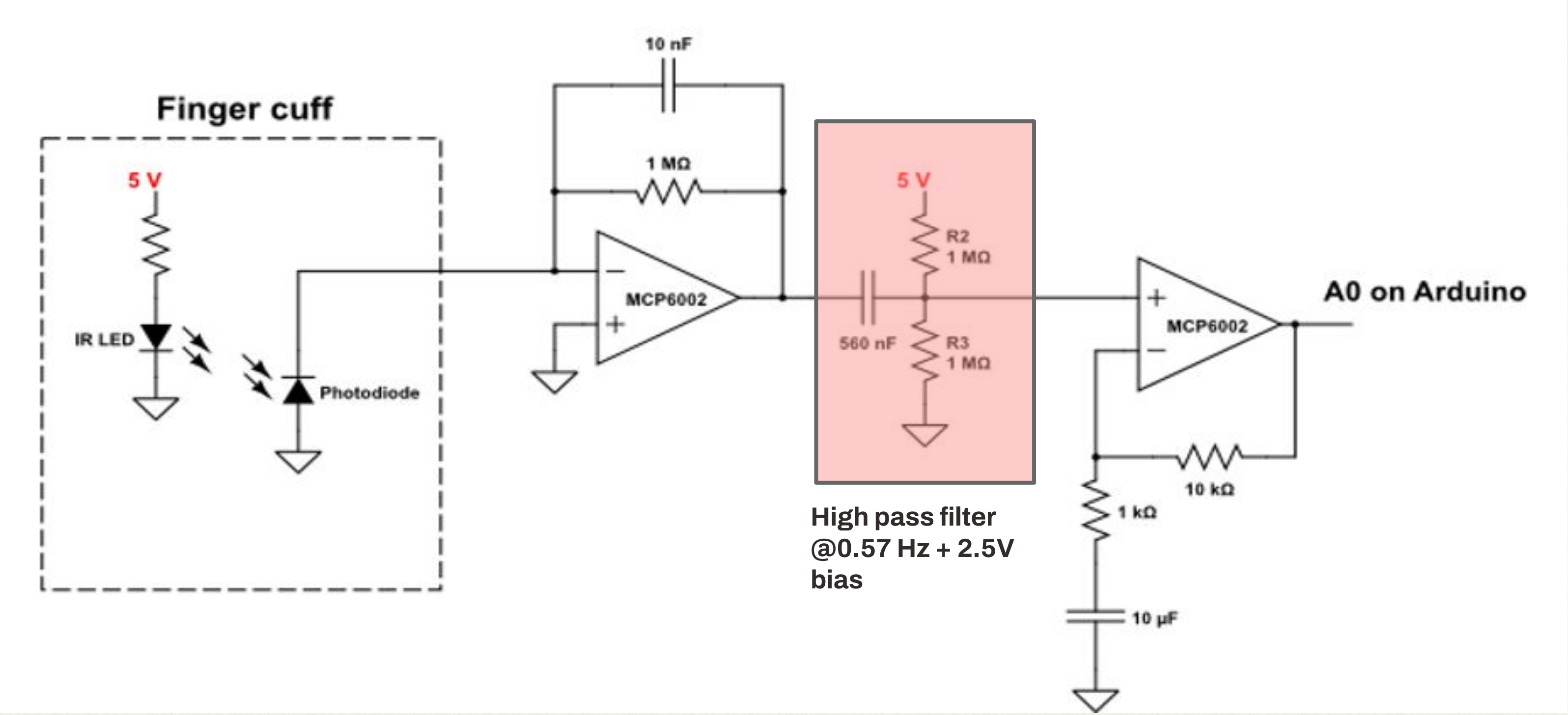
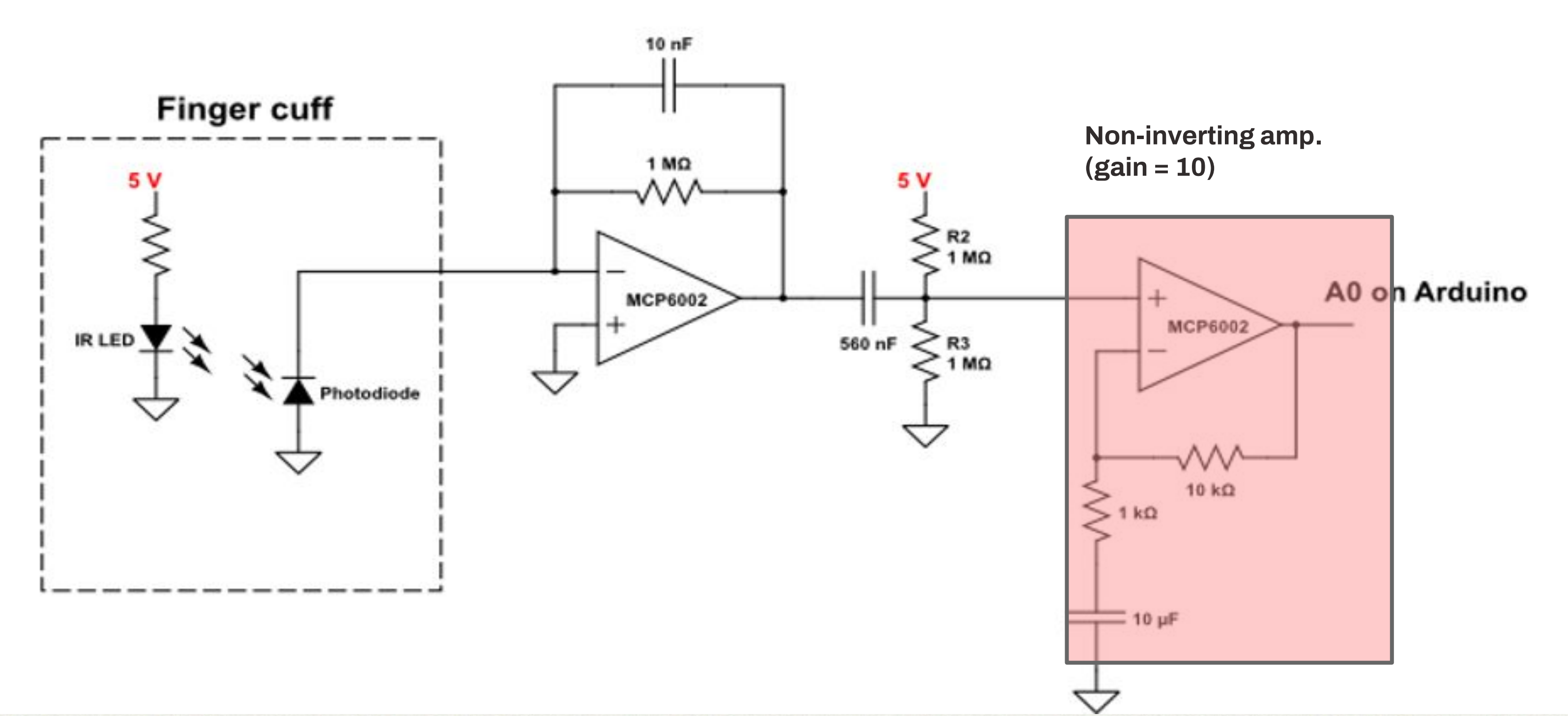


High pass filter @0.57 Hz + 2.5V bias

Electrical Design



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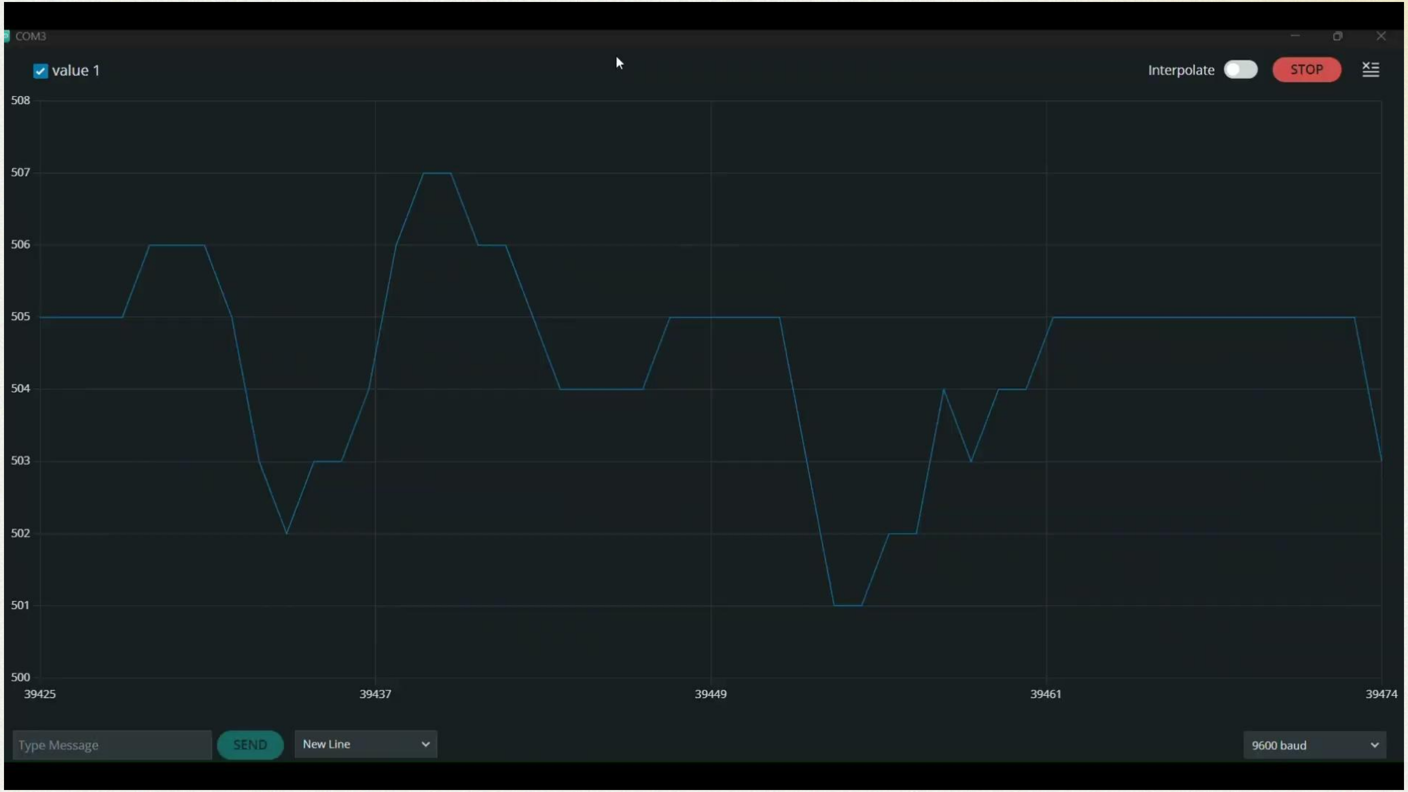
Code - Peak Detection

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    }
    peakIndices[peakCount++] = i;
    lastPeak = i;
    if (peakCount >= 20) break;
```

Code - BPM and LCD

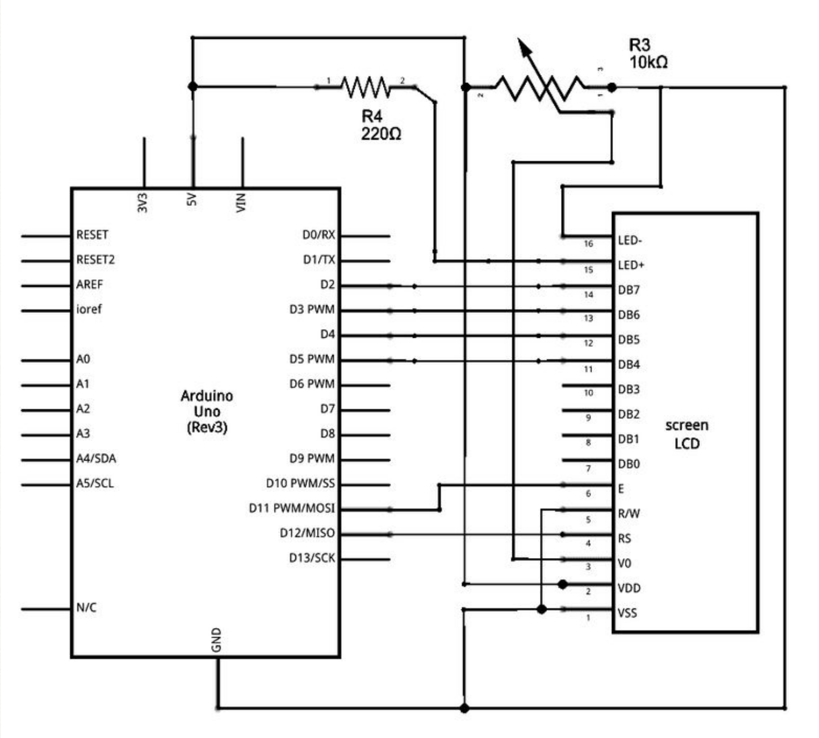
```
float calculateBPM() {
  if (peakCount < 2) return -1;

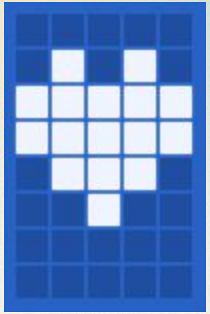
  int numIntervals = peakCount - 1;
  float intervals[19];
  for (int i = 0; i < numIntervals; i++) {
    intervals[i] = peakIndices[i + 1] - peakIndices[i];
  }

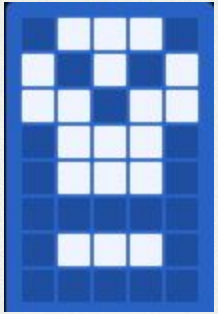
  for (int i = 1; i < numIntervals; i++) {
    float key = intervals[i];
    int j = i - 1;
    while (j >= 0 && intervals[j] > key) {
      intervals[j + 1] = intervals[j];
      j--;
    }
    intervals[j + 1] = key;
  }

  float medianInterval;
  if (numIntervals % 2 == 1) {
    medianInterval = intervals[numIntervals / 2];
  } else {
    medianInterval = (intervals[numIntervals / 2 - 1]
    + intervals[numIntervals / 2]) / 2.0;
  }

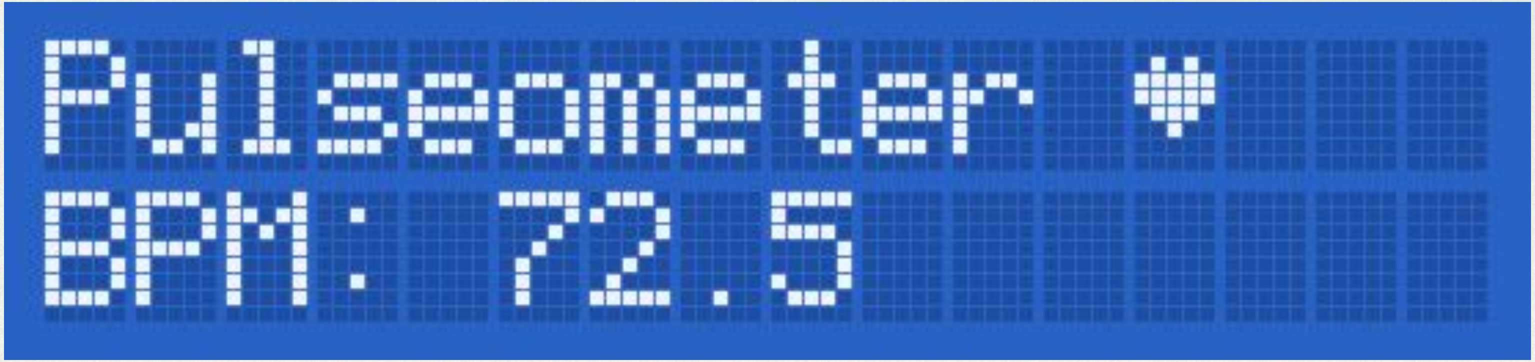
  float fs = 1000.0 / SAMPLE_INTERVAL;
  return 60.0 / (medianInterval / fs);
}
```







LCD Screen



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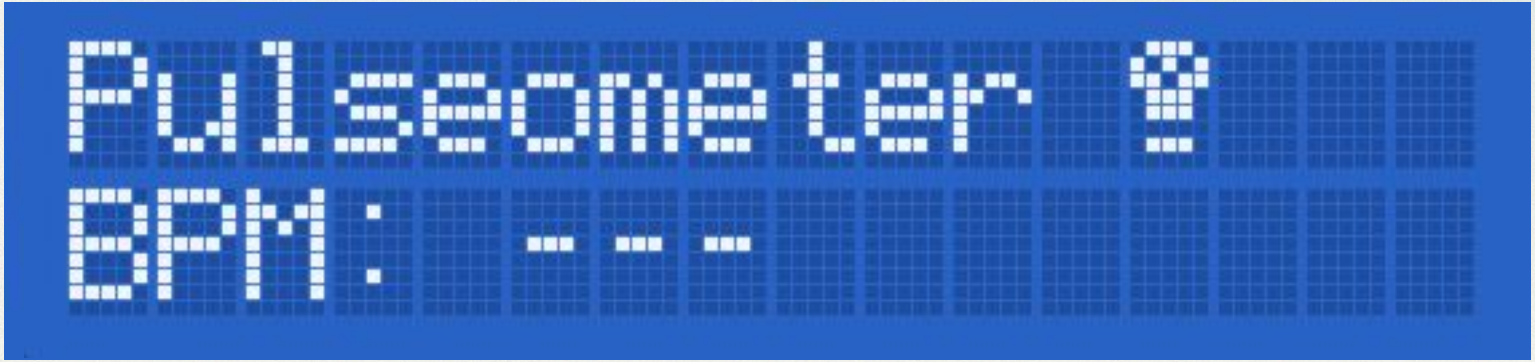

Mechanical Design - prototyping

Semester 1:

Reflective PPG

“Watch” design

Very poor signal

Very poor stability

Mechanical Design - prototyping

Jan-Feb:

Transmittive PPG

“Finger box” design

Poor signal

Poor stability

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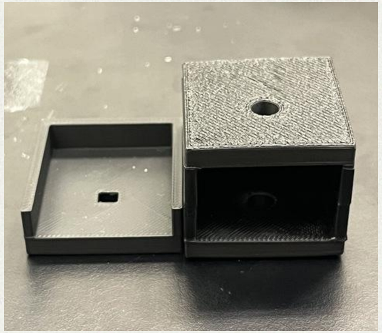

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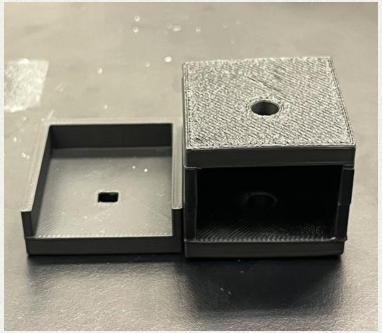

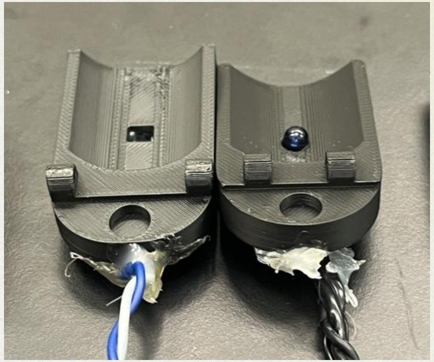

Feb-Mar:

Transmittive PPG

Initial finger clip design

Better stability

Better optical isolation

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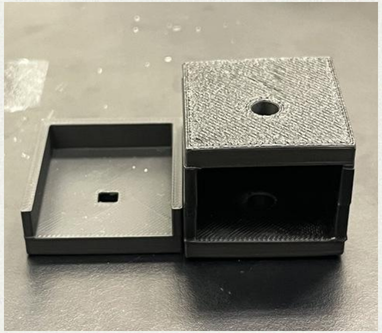

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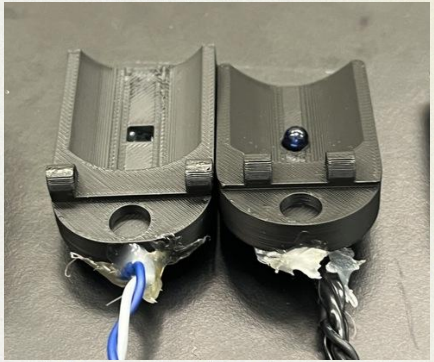

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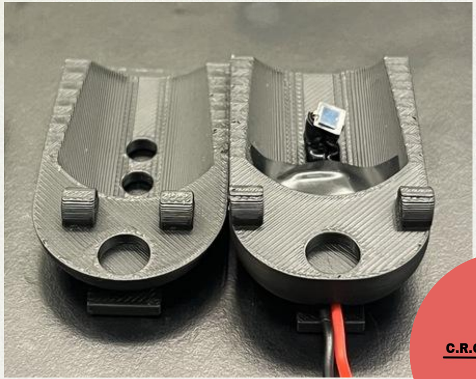


Mar-Apr:

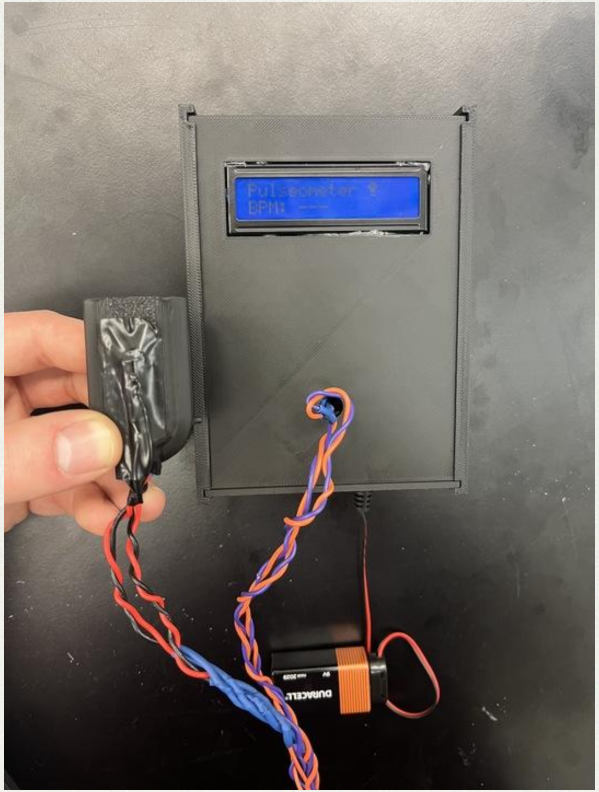
Transmittive PPG

Potential future finger clip design

More ergonomic

Sunken photodiode

Mechanical Design - Final Product



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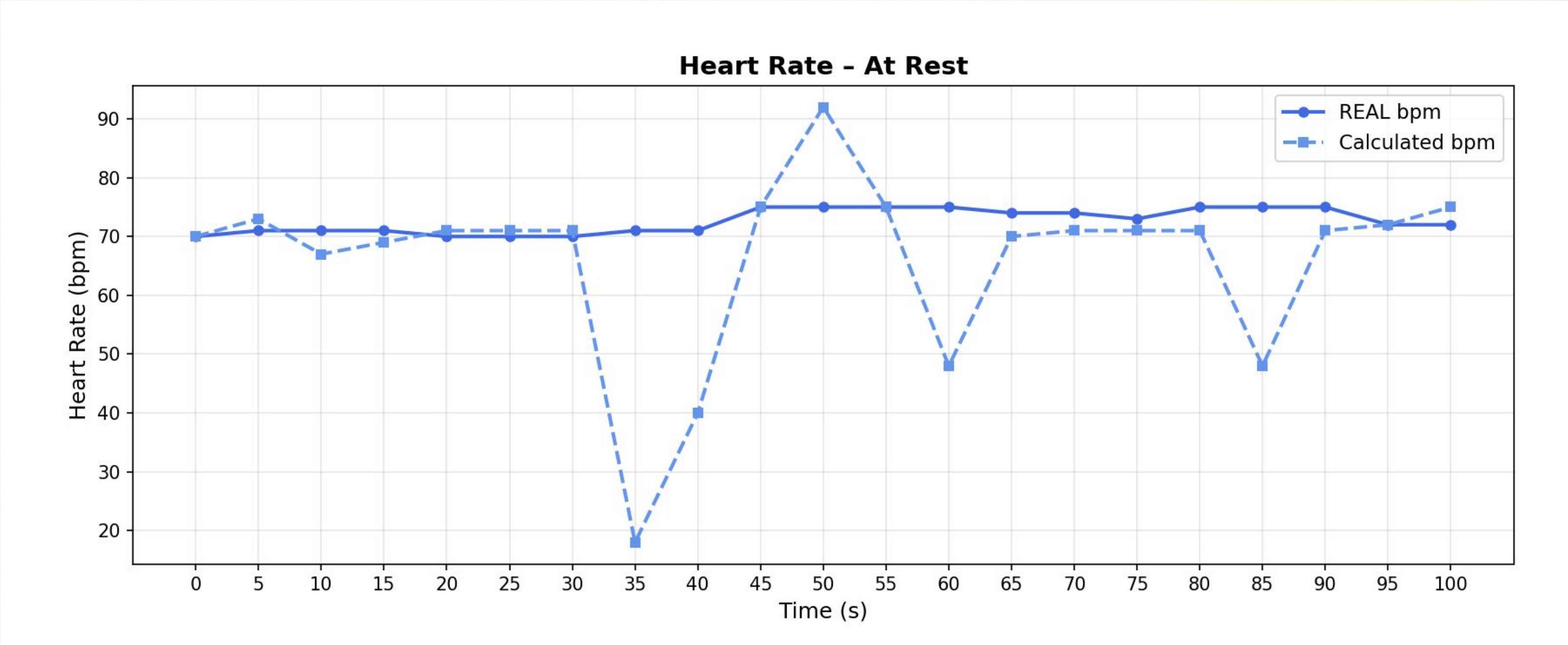
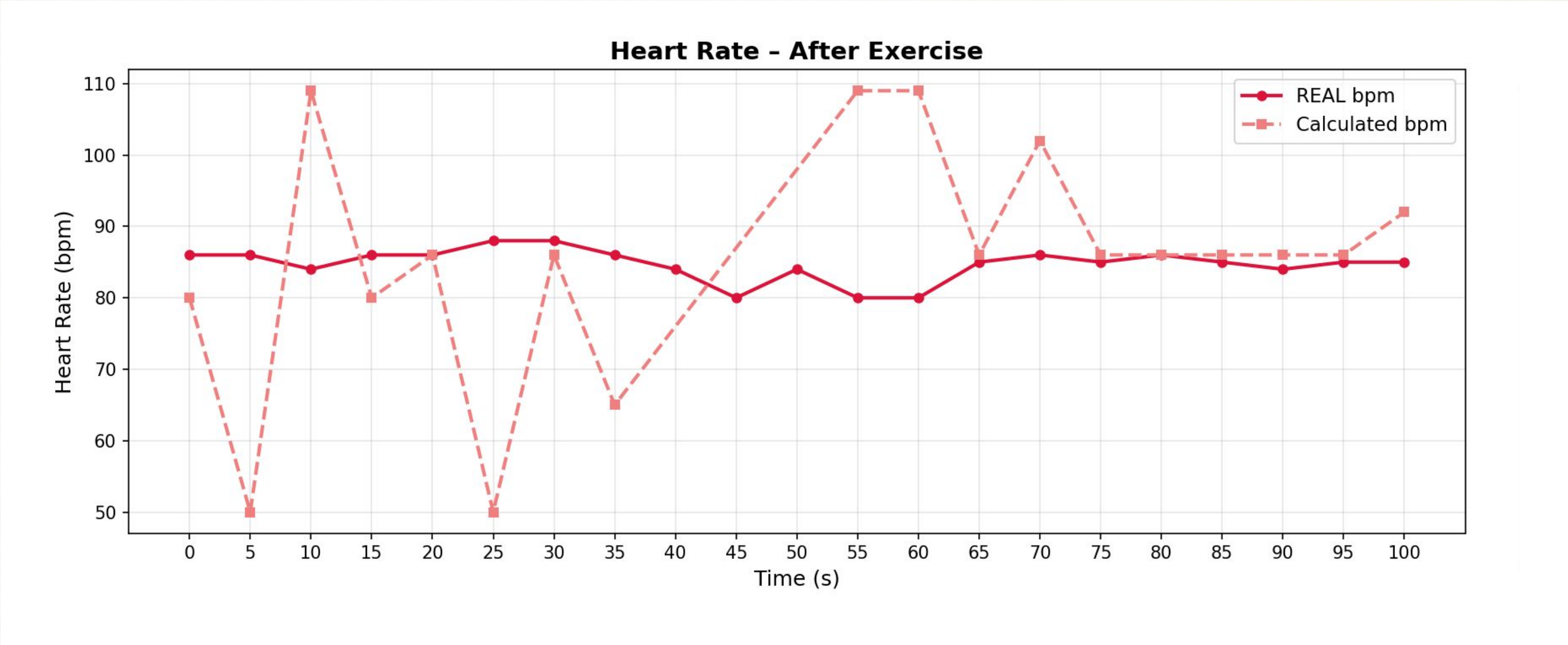
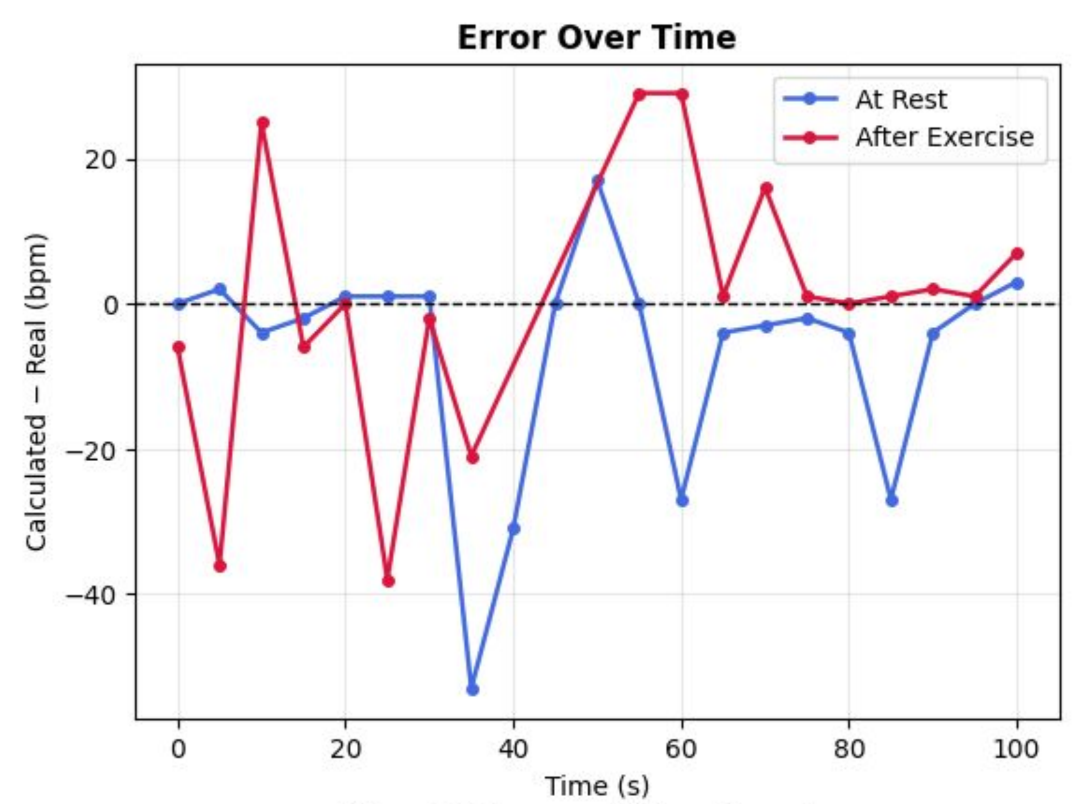
Performance Benchmarking



Performance Benchmarking



Performance Benchmarking - Error



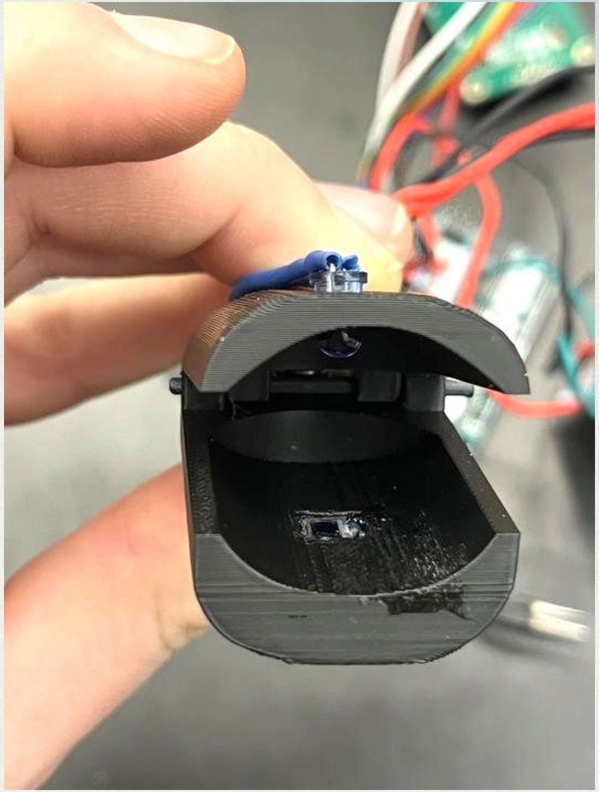
At rest:
MAE = 8.9 bpm
After exercise:
MAE = 12.3 bpm

Performance of device entirely dependent on grip / orientation of finger

Safety & User Considerations

Safety:
- LED bright enough for signal, without overheating
- 3D printed design is ergonomic and comfortable

Future Improvements:
- LED pokes out into finger
- LED wiring visible and slightly flimsy
- 3D printing plastic should be replaced with injection mold plastic / foam
- Balancing out clip strength with signal decay



Final Remarks - How do we compare?

Competitors

Function - variety of options (watches and fingers clips)

Accuracy - MAE is 0 bpm (since this was the benchmark)

Price - average $52 - $522

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Function - only finger clip option

Accuracy - rest MAE is 8.9 bpm & exercising MAE is 12.3 bpm

Price - $13.40 - $23.40

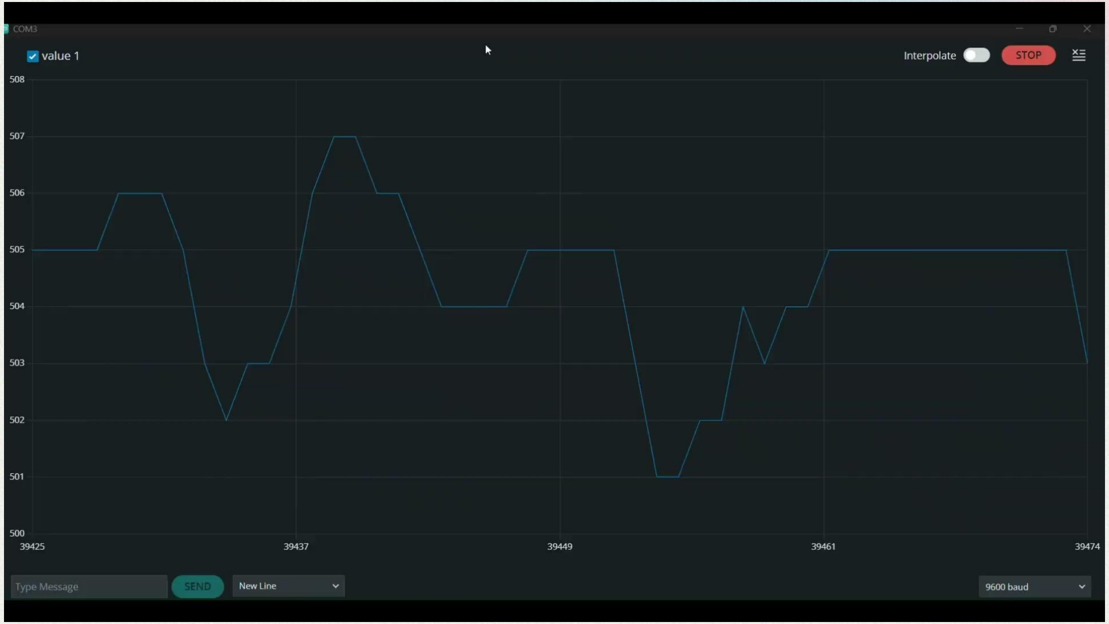
Final Remarks - Strength & Weakness

Strengths

- Waveform detection
- Affordability

Weaknesses

- Peak Detection
- Ergonomics
- Reliability



Final Remarks - C.R.O.I Pulometer 2

Pulsometer 2 Added Features

- Bluetooth
- Oximeter Readings

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C.R.O.I

Chinedum Osuji -
Lead Strategic Consultant,
CROI technologies

John Crocker -
Lead Technical Consultant,
CROI technologies

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Q&A

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